Football's Tactics Problem Is a Power Problem

Project Underdog: ten pre-registered rounds, 28 tests, two surviving tactics, and how one club could prove them in a season

Autonomous research run, 11–12 October 2026. Every round was pre-registered in git before its data were analysed (`PREREGISTRATION*.md`). Deviations are logged in `DEVIATIONS.md`.

Abstract

We set out to find a tactic that lets a weaker football team, such as Brentford, beat stronger ones. Over nine rounds we pre-registered and tested 28 hypotheses. The data were event data for 3,954 matches across four independent populations (StatsBomb big-five 2015/16, Wyscout big-five 2017/18, unseen StatsBomb WSL/NWSL/ISL) and closing odds for 45,825 matches.

Almost every candidate died. The casualties include:

- in-game style knobs and coaching changes;
- optimised in-match policies (whose in-sample gains were 3× the honest ones);
- box-score style traits and style matchups against the betting market, including a machine-learning search;
- opponent "weak spots";
- defensive cone density (strong in-sample, p = 0.026; failed all three confirmations);
- "chase earlier" timing;
- corner targets and long throws, which passed confirmation on the letter and then dissolved under same-location and within-team controls;
- referee adaptation.

Two patterns survived everything.

1. **Set pieces are where quality matters least.** Set-piece xG elasticity to team strength is 0.41/0.28/0.72, against 1.09/0.89/1.02 for open play, in three independent datasets.
2. **"Break after they commit."** At the same regain spot, score and minute, a regain made right after the opponent had pushed into your box yields 1.4–1.7× the counter xG. This held in all four populations, every 95% CI above 1.

Round 10 explains the graveyard, and that is our main contribution. Football tactics live in a **low-power regime**:

- detecting a corner routine 20% better needs 15 seasons of one club's corners;
- a match-level tactic worth +0.1 points per match (≈4 points a season, which is large) needs 57 club-seasons of results.

Every club's history, and every observational "insight" mined from it, is drowned in noise and selection. That is why football never had its fourth-down or three-point moment.

The way out is the clinical-trial playbook: randomise, and measure a validated surrogate endpoint. For transitions, "reached the final third within 10 seconds of winning the ball" captures 66% of regain xG. It predicts a team's future chance creation almost as well as xG itself (r = 0.27 vs 0.31), and it cuts

the sample needed **tenfold**. A 20% better transition rule becomes detectable in **0.7 club-seasons**. The "break after they commit" effect is already visible in that surrogate (odds ratio 1.47 [1.36, 1.59]; unseen leagues 1.26 [1.08, 1.47]). A general transition rule can be tested by one club in under a season. The specific 'break after they commit' rule needs about two seasons for one club, or under one for a three-club consortium. We give the protocol.

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1. The question

The brief was to treat football as a sequential decision problem and look for a tactical policy, not a prediction model, that lets a weaker squad consistently outperform stronger ones. "Rediscovering" set pieces and direct play would not count; the goal was something materially better, with credible counterfactual validation.

Two traps were clear from the start, and the design is built around them.

1. **Historical data show what teams did, not what would have happened otherwise.** Style and outcomes are entangled. When the opponent attacks, our defensive actions are deep *because* they are attacking.
2. **An optimiser will exploit its own simulator.** A policy tuned on a fitted model always looks good when it is scored on the same model.

So we pre-registered the hypotheses, kept the 2015/16 Premier League out of model fitting, and evaluated every optimised policy on a fit it had not seen.

2. Data

Source	Coverage	Used for
StatsBomb Open Data	1,551 matches, 2015/16: full EPL, La Liga, Serie A, Ligue 1, plus 34 Bundesliga. 5.4M events with xG	style, segments, coaching decisions, chance channels
Wyscout public dataset (Pappalardo et al., 2019)	1,826 matches, 2017/18, big five	out-of-season, out-of-provider replication
StatsBomb Open Data (unseen)	577 matches: WSL 2018/19–2020/21, NWSL 2023, ISL 2021/22	confirmation only (rounds 5–6)
football-data.co.uk	45,825 big-five matches 2000/01–2025/26, closing odds + shots	market strength, P4, 2025/26 season simulation

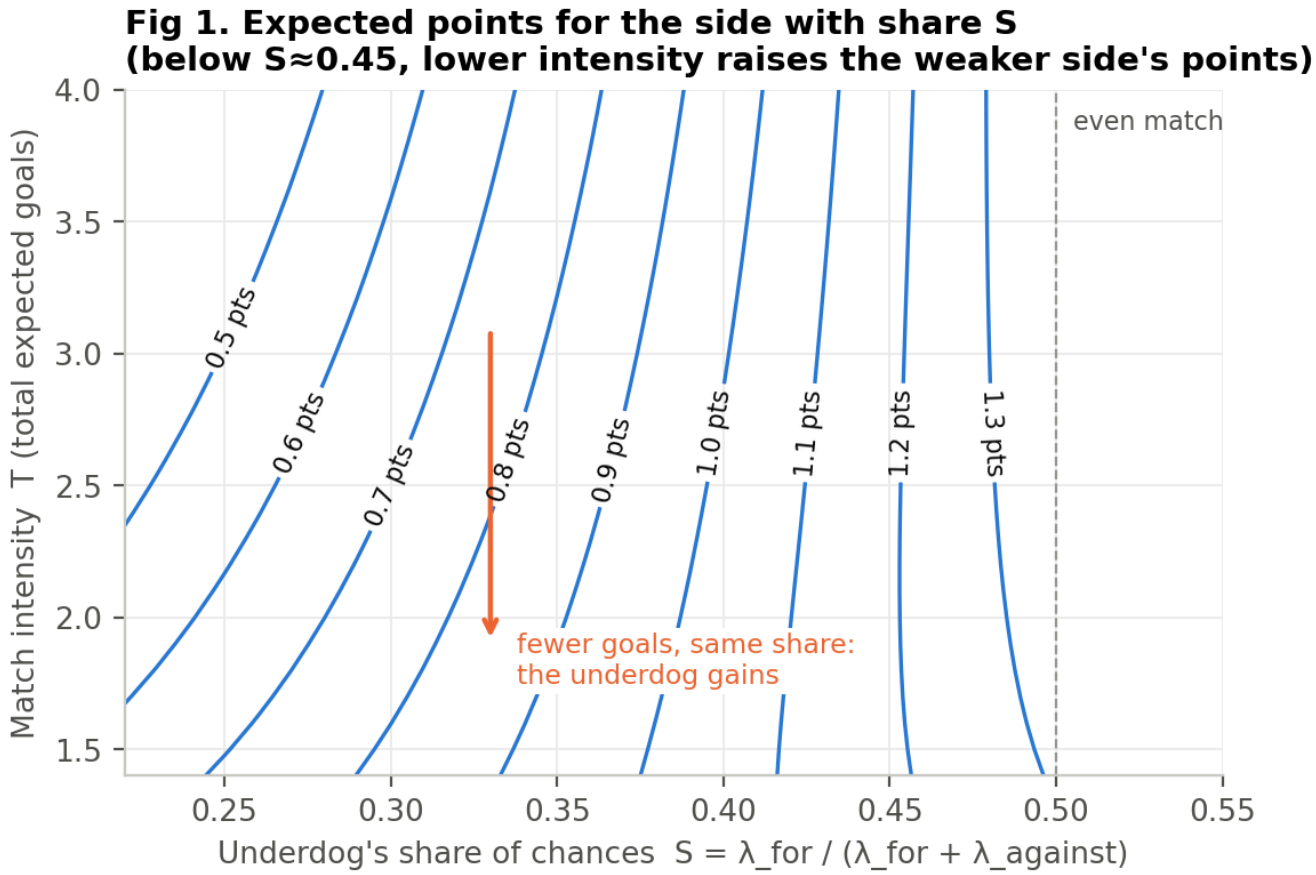
Market strength. For every match we de-vig the closing 1X2 odds (Pinnacle closing where available) and invert them, together with the over/under 2.5 price when available, into Poisson goal rates (λ_home, λ_away) with a Dixon–Coles correction. This is the strength baseline throughout. It is demanding: anything the market already knows is not an edge.

Segments. Each match is cut into stretches of constant game state. Cuts fall at 15-minute marks, every goal and every red card, giving 27,570 team-segments. For each one we record xG for and against, and four style descriptors: defensive-action height, pressures per opponent pass, long-pass share and pass share.

3. Framework: the underdog's exchange rate

Write a team's chance production as λ_for and λ_against. Any tactic moves the match along two axes: **intensity** T = λ_for + λ_against, and **share** S = λ_for / T. Figure 1 shows expected points for the side with

share S. Below S ≈ 0.45 the iso-point lines tilt, so at a fixed share **fewer total goals raise the weaker side's points.** More draws, and more noise relative to quality.

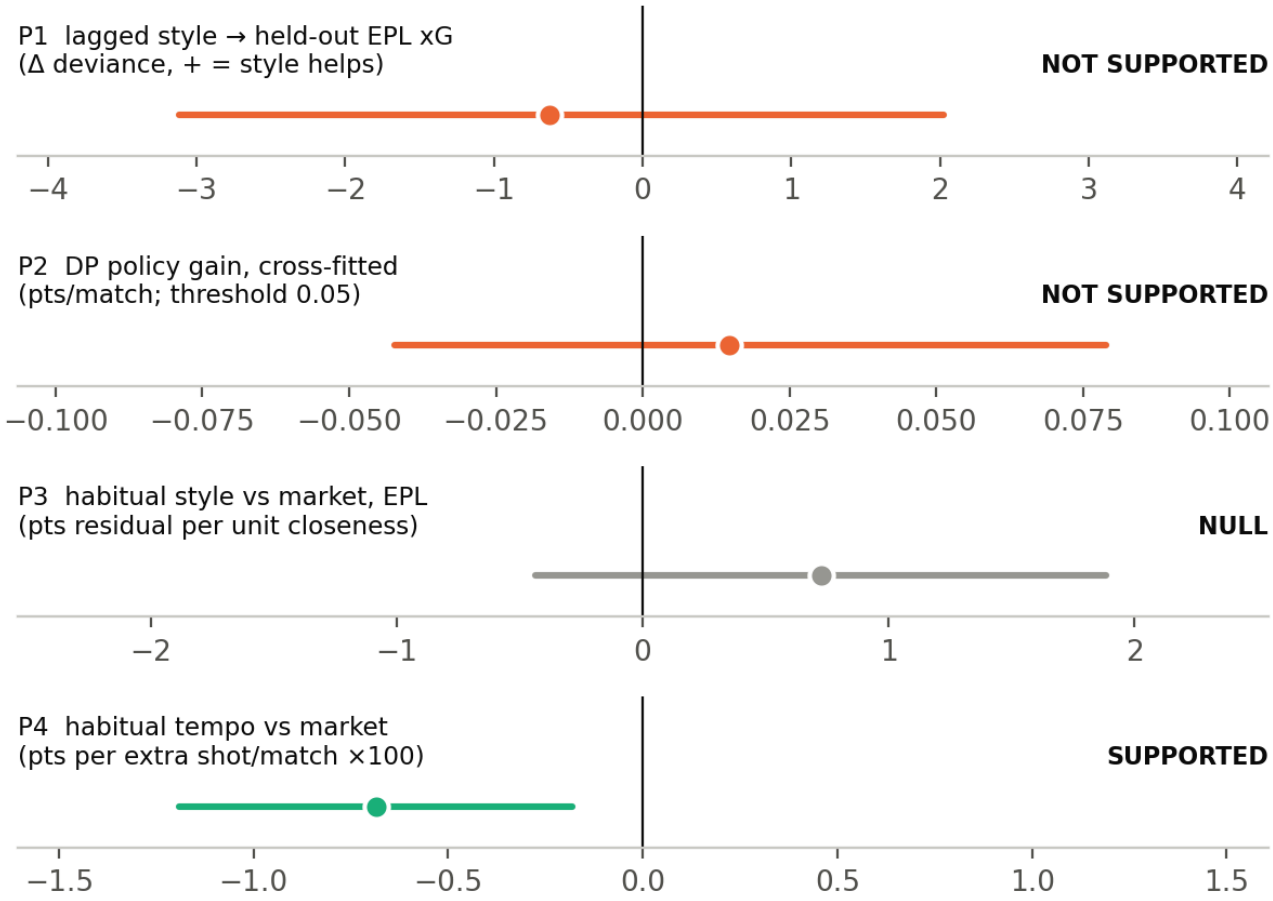


The exchange rate is concrete. For a 30%-share underdog in a 2.7-goal match, cutting total intensity by one goal is worth +0.10 points/match, the same as gaining 3.4 percentage points of share. For a 40%-share side the same cut is worth only +0.045. This is the theoretical case for "shrinking the game". The empirical question is whether any lever moves T without giving back more than that in S.

The in-match decision problem is solved exactly. The state is (minute, goal difference), the actions are style modes with per-minute scoring rates, and the terminal utility is league points (3/1/0) or any other utility. Backward induction gives the optimal policy, and a forward pass gives the outcome distribution (`src/dp.py`).

4. Pre-registered results

Fig 2. Pre-registered tests (point estimate, 95% CI). P5 shown in Table 3



	Hypothesis (registered direction)	Result	Verdict
P1	Lagged style improves held-out EPL xG prediction	Δdev −0.62, CI [−3.12, +2.02]	**Not supported**
P2	DP policy gains ≥ 0.05 pts/match for a heavy underdog, robust to refitting	cross-fitted +0.015, CI [−0.042, +0.079]; positive in 70% of refits	**Not supported**
P3	Habitual style near the P2 prescription beats the market (EPL)	+0.72, CI [−0.44, +1.88]; n = 330	**Null**
P4	Habitual tempo negatively predicts underdog residual vs market	−0.0069 pts/shot, CI [−0.0119, −0.0018], p = 0.008; n = 36,374	**Supported (modest)**
P5	Title-maximising policy is more variance-seeking than points-maximising	title probability flat in κ (Table 3)	**Not supported**

P1. With style lagged by one segment (the primary specification, fixed in DEVIATIONS #1 before results were seen), style adds nothing out of sample. The contemporaneous model "improves" deviance by 7.1% (CI [+5.5%, +8.9%]), but its signs show it is mechanical. The opponent's defensive height is the strongest "predictor" of our xG (−0.28 log-points per SD) because deep opponents are a *consequence* of our attacking. A naive analysis would have reported that result as a discovery.

P2. The registered frontier, the lagged-style GLM with 27 style actions within ±1 SD of observed support, recommends pressing high and going long. Evaluated three ways:

Evaluation	Gain (pts/match)
Naive: optimise and evaluate on the same fit	+0.027
Literal registered criterion: refit on 60 bootstrap resamples, re-optimise	mean +0.051, **100%** of resamples > 0
Cross-fit: optimise on resample *b*, evaluate on independent resample *b'*	**+0.015, CI [−0.042, +0.079], 70% > 0**

The literal criterion is met, but it is trivially satisfiable: an optimiser that picks the best of 27 noisy actions always finds a positive gain on its own fit. We regard this as a flaw in our pre-registration wording and report the cross-fitted number as the real result. This is the "simulator exploitation" failure mode the brief warned about, measured. In-sample gains overstate the honest gain by about 3×.

P4. Underdogs whose previous 10 league matches featured more total shots underperform their closing-market expectation. The sign is negative in every league (EPL −0.0075, Bundesliga −0.0085, Ligue 1 −0.0084, Serie A −0.0076, La Liga −0.0031). It survives controlling for the team's own shot share (−0.0065, p = 0.012). It is strongest in 2005–14 (−0.0113, p = 0.003) and indistinguishable from zero since 2015 (−0.0033, p = 0.34), consistent with a market that has learned. It is null for heavy underdogs (p_win < 0.25: −0.0025, CI [−0.0088, +0.0038]). At one SD of habitual tempo (2.3 shots) the effect is ~0.016 points/match, about 0.6 points/season.

Is P4 the variance mechanism? Probably not. If the market ignored variance entirely, the residual would equal the Poisson-theory effect of tempo. Pooled, theory and data agree (−0.0052 vs −0.0069 per shot), but by strength band they diverge (`results/theory_vs_p4.json`):

Underdog p_win	n	Empirical	Variance theory
< 0.20	14,324	−0.0018 (0.0036)	−0.0104
0.20–0.30	14,442	−0.0060 (0.0042)	−0.0035
0.30–0.45	7,608	−0.0177 (0.0062)	+0.0014

Theory says heavy underdogs should gain most from low tempo; the mispricing is instead concentrated in *mild* underdogs, where theory predicts nothing. The market appears to price the variance effect for long shots and to overrate high-tempo "chaos" teams that are only slightly weaker. That is a market-efficiency finding, not a tactical one.

5. Post-hoc analyses (hypothesis-generating)

5.1 Coaching decisions barely move chance rates

After P1 failed, we turned to the levers coaches actually pull in-game. There were 6,789 substitution and formation-change decisions, each scored by the change in the team's positional attacking load (`src/tactical_changes.py`). Post-decision xG over the next ≤15 minutes (stopping at the next goal) was compared with a strength + state baseline, controlling for pre-decision momentum and the opponent's simultaneous changes.

Decision (vs like-for-like)	Own xG	xG conceded
Attacking (n = 539)	×1.04 (SE 0.06)	×1.03 (SE 0.06)

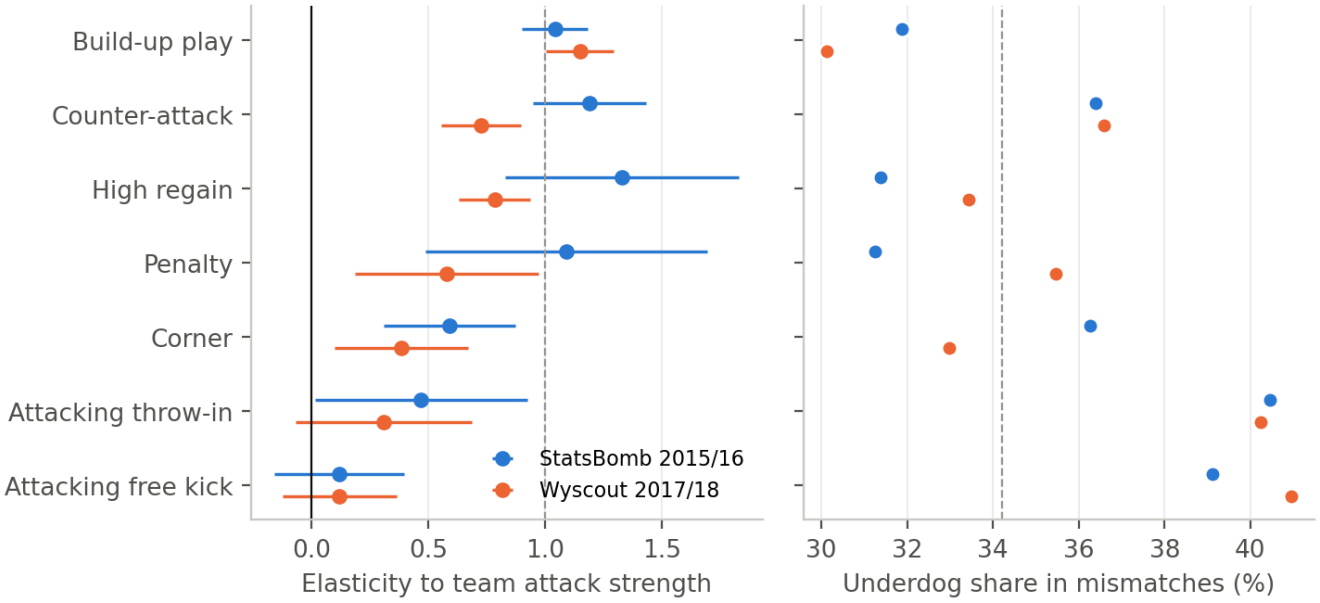
Decision (vs like-for-like)	Own xG	xG conceded
Defensive (n = 488)	×1.13 (SE 0.07)	×1.03 (SE 0.07)

Neither kind of change detectably moves either rate. The DP built on these effects shows a cross-fitted gain of +0.035, CI [−0.075, +0.185]. What does move rates is game state itself: trailing teams' xG rises steeply late in matches (late × trailing: +0.32 log-points per 90 min). Every team already responds to the score; deliberate changes add little on top that we can detect.

5.2 Where quality matters less: chance channels

We split every possession into channels by origin: build-up, counter-attack (regain → shot ≤ 15 s, or tagged), high regain (won in the final third), corner, attacking free kick, attacking throw-in and penalty. Restarts count as set pieces only if any shot comes within 20 s. Each channel's team-match xG was then regressed on market attacking strength.

Fig 3. Where quality matters less: set pieces are flat, build-up is steep



Channel	Elasticity: StatsBomb 15/16	Elasticity: Wyscout 17/18	Underdog share in mismatches (SB / WY)
Build-up	1.04 (0.07)	1.15 (0.07)	31.9% / 30.1%
Counter	1.19 (0.12)	0.73 (0.09)	36.4% / 36.6%
Corner	0.59 (0.14)	0.39 (0.15)	36.3% / 33.0%
Attacking throw-in	0.47 (0.23)	0.31 (0.19)	40.5% / 40.2%
Attacking free kick	0.12 (0.14)	0.12 (0.12)	39.1% / 41.0%
All open play	**1.09**	**0.89**	**33.0% / 33.1%**
All set pieces	**0.41**	**0.28**	**38.0% / 36.9%**

Elasticity 1 means a team with 10% more attacking strength creates 10% more xG in that channel. The Wyscout elasticities are uniformly lower because its xG is our location-only model (AUC 0.78 vs StatsBomb's 0.815), which compresses chance-quality differences. Compare ratios, not levels.

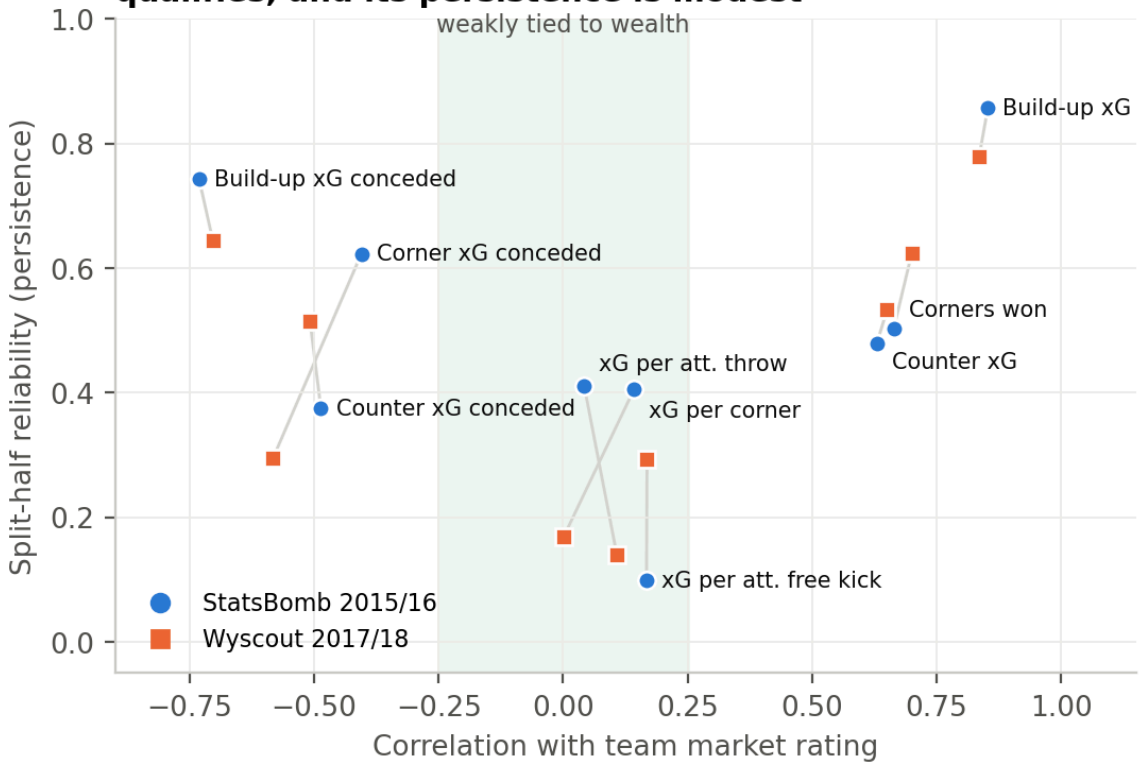
What replicates: **set pieces are roughly a third as quality-sensitive as open play**, in both datasets, with different xG models and two seasons apart. **Build-up play is where underdogs lose most** (30–32% share). Underdogs also keep a relatively large share of counter-attack xG (36–37%) in both datasets.

What does not replicate: the counter-attack *elasticity* (1.19 vs 0.73). We drew an early "counters are not an equaliser" conclusion from the first dataset and withdrew it.

5.3 The cheap-skill map

A useful underdog lever is a team skill that is **persistent** (a real ability, not noise) and **not bought** (uncorrelated with team strength). We computed split-half reliability (odd vs even matches, Spearman–Brown corrected) and the correlation with the team's market rating for each metric.

Fig 4. Cheap skills (persistent, not bought): only set-piece efficiency qualifies, and its persistence is modest



Most persistent metrics are wealth-bound: build-up xG for and against have reliability 0.65–0.86 and |r| = 0.70–0.85 with rating. Only set-piece *efficiency* sits in the cheap region. xG per corner has r = 0.14/0.00 with rating and xG per attacking throw r = 0.04/0.11. Their persistence is modest, though: 0.41 for both in StatsBomb, and 0.17/0.14 in Wyscout. Corner *defence* is more persistent (0.62/0.30) but is partly wealth-linked (r = −0.41/−0.58). This is Brentford's existing edge. The data say it is real but small, and noisy enough that a single season cannot confirm a team has it. It is also already priced: a team's habitual net set-piece xG does not predict its points against the closing market (StatsBomb +0.07 ± 0.22 per xG/match; Wyscout −0.13 ± 0.24; `results/setpiece_market.json`). A club can still *build* the skill cheaply, but it cannot hide it.

6. Brentford, 2025/26

We fitted attack/defence ratings to the 2025/26 Premier League closing-market rates and simulated 200,000 seasons. Non-Brentford matches use Dixon–Coles Poisson scorelines; Brentford matches use the DP outcome distribution under each intervention. Baseline Brentford: 50.2 expected points (actual: 53), 11th on average.

Interventions are restricted to observed support:

- **Set pieces to the 90th percentile** of *true* team skill (observed SD × √reliability). That is +0.058 xG/match for and −0.052 against with StatsBomb reliabilities, or +0.036/−0.036 with Wyscout's.
- **Tempo menu** {×0.91, ×1, ×1.09} on both teams' rates at a fixed share. ±9% is one SD of the between-team spread in habitual tempo (P4 data). The DP chooses by (minute, score). This is an **upper bound**: P1, P2 and §5.1 found no in-game lever that does this reliably.

Fig 6. Brentford: what the evidence-backed playbook is worth over 38 matches

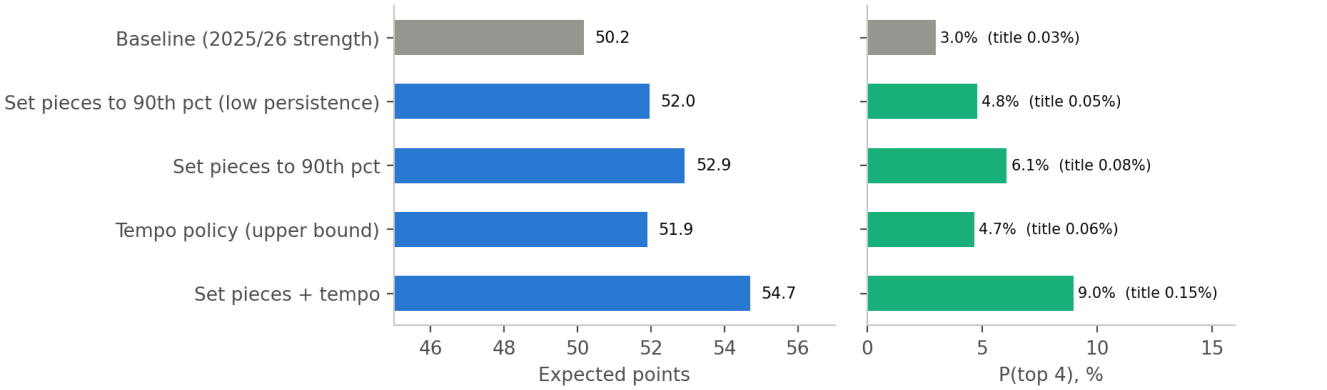


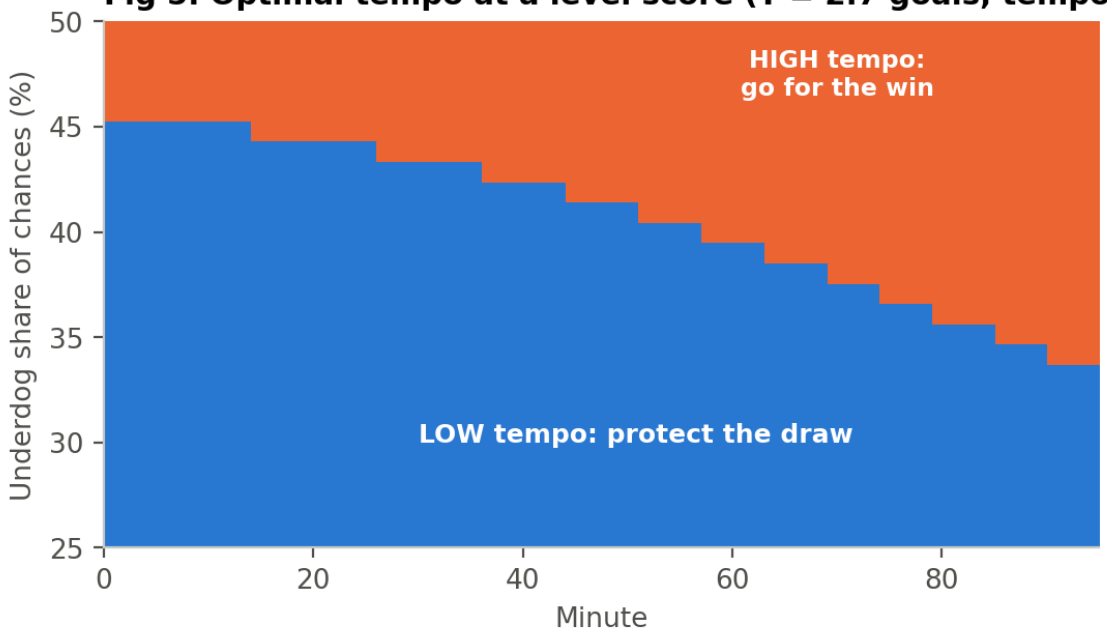
Table 3. Brentford season outcomes (200k seasons; risk-dial rows 300k, common random numbers).

Scenario	E[pts]	P(top 4)	P(title)
Baseline	50.2	3.0%	0.03%
Set pieces p90 (low persistence)	52.0	4.8%	0.05%
Set pieces p90	52.9	6.1%	0.08%
Tempo policy (upper bound)	51.9	4.7%	0.06%
Set pieces + tempo	54.7	9.0%	0.15%
… + risk dial κ = 1, 3, 8 (P5)	54.7	9.0%	0.15%
Reference: +10% attacking quality	*53.8*	*7.5%*	*0.11%*
Reference: +40% attacking quality	*65.6*	*48.6%*	*3.6%*

The full playbook is worth about the same as a 10–12% improvement in attacking quality. P5 was also tested at contender strength (+40% and +60% attack; title probability 5.7% and 25.9%). The risk dial $\kappa \in \{0, 1, 3, 8\}$, which values wins above draws, never raised title probability and slightly lowered it in both cases. Over a 38-match season, maximising expected points is title-optimal.

When to open up. Figure 5 reads the "drawing at 70 minutes" question straight off the value function. With a ±9% tempo menu and level scores, a side with under ~45% of the chances should play low-tempo from kick-off. The share at which it should switch to high tempo falls steadily through the match, to about 34% by full time.

Fig 5. Optimal tempo at a level score (T = 2.7 goals, tempo menu ±9%)



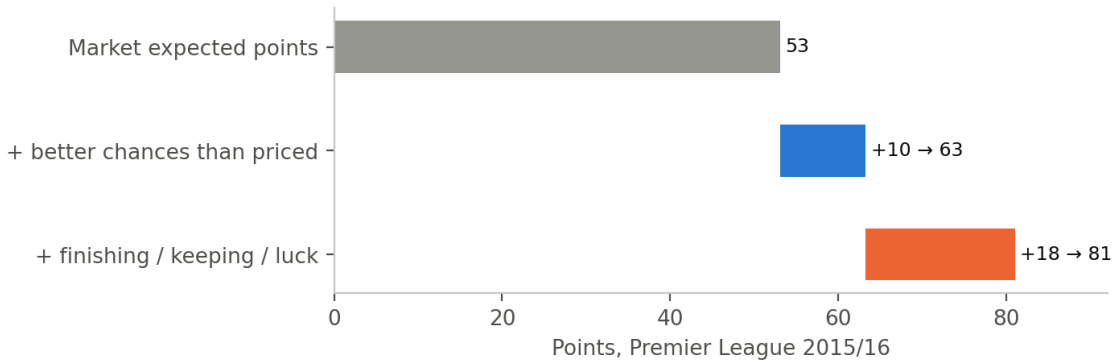
For Brentford vs Manchester City (2025/26 strengths), the expected points and policy are:

	Home (λ 1.13 vs 1.95)	Away (λ 0.93 vs 2.37)
Baseline E[pts]	0.872	0.566
DP tempo policy	0.910	0.606
Set pieces + DP tempo	0.975	0.659
Level at 70'	protect (low)	protect (low)
Level at 85'	**go (high)**	protect (low)
Trailing by 1	go (high) from kick-off	go (high) from the 45th minute

So the answer to the brief's question is: **at home, keep the draw until about the 85th minute, then go for it; away, keep the point.** All of this is conditional on the tempo lever existing.

7. Leicester 2015/16: the one underdog champion

Fig 7. Leicester 2015/16: two thirds of the miracle sat above their chances



Leicester's market-expected points were 53.1; they finished on 81. Their xG-implied expected points were 63.3. So roughly 10 points came from creating better chances than the market priced, and 18 from finishing, goalkeeping and fortune above their own chance quality. Their style did not shrink the game. They ranked **1st in tempo** (2.96 xG per match, both teams), **1st in counter-attack share** (24% of their xG), 18th in

possession, 4th in long-ball share. They also had by far the most penalty xG (10.2). They opened matches up, which is the opposite of the variance prescription. The rare underdog title in our data looks like underpriced talent (Vardy, Mahrez, Kanté) plus a large positive draw from the noise. We found no reusable structural trick in it.

8. Part II: five more rounds of searching

Round 1 left one robust but unoriginal finding: set pieces. The brief asked for more, so the search continued. Each round was registered in git before its data were analysed, with an explicit confirmation step.

Fig 8. 28 pre-registered tests, nine rounds: 7 held

R1	P1 style predicts next-segment xG	✗
R1	P2 in-match policy ≥0.05 pts (cross-fit)	✗
R1	P3 habitual style beats market	✗
R1	P4 high-tempo underdogs lose vs market	✓ **held**
R1	P5 risk-seeking raises title odds	✗
R2	3 box-score traits/matchups (BH-selected)	✗
R2	ML on 22 style traits beats market	✗
R3	Opp. commitment improves regain model	✗
R3	Counter value falls if opp. was deep	✗
R3	Top 20% regains carry ≥40% of xG	✓ **held**
R3	Concession profiles persistent	✗
R3	Matchup model beats main effects*	✓ **held**
R4	Structure predicts goals beyond xG	✗
R4	Cone density persistent, not bought	✓ **held**
R4	Over-performance vs xG repeats	✓ **held**
R5	Cone density → fewer goals (prospective)	✗
R5	… on unseen leagues	✗
R5	… shot level, unseen	✗
R6	Set pieces flatter than open play (unseen)	✓ **held**
R6	Underdog keeps more set-piece xG (unseen)	✗
R6	Counter value rises if opp. was deep (unseen)	✓ **held**
R7	Trailing teams escalate late	✗
R7	Teams chase too late ('pull the goalie')	✗
R8	Corners to penalty spot beat near post†	✗
R8	Long throws into box beat short†	✗
R9	Strict referees hurt underdogs	✗
R9	… especially foul-heavy underdogs	✗
R9	Referee strictness shifts total goals	✗

Round 2: a discovery engine with an out-of-time lock

- **Search.** 120 tests on 2005–2017 (19,605 underdog matches): 10 habitual traits of the underdog, 10 of the favourite, and all 100 underdog × favourite matchups. The outcome was goal difference against the closing market. The P4 tempo effect reappeared as a positive control ($p = 0.008$), so the pipeline is sound.
- **Shortlist.** Benjamini–Hochberg at FDR 10% kept three effects: favourite shot volume, favourite shots conceded, and *card-prone underdogs × foul-drawing favourites*.
- **Confirmation.** On the locked 2017–2026 seasons (15,818 matches) **none confirmed**. All three shrank by 85–95% toward zero.
- **ML arm.** A LightGBM model on all 22 traits did no better than the market on the holdout (MSE gain −0.0011, CI [−0.0024, +0.0002]).

Conclusion: since 2017 the market prices all box-score style information.

Round 3: when is a stronger team vulnerable, and do teams have weak spots?

- **Regain data.** Every open-play regain in StatsBomb 2015/16 (110,541) and Wyscout 2017/18 (430,975), scored by the xG of the following 20 seconds.
- **H3a-1 failed.** Features describing the opponent's preceding possession did not improve held-out prediction once game state was known.
- **H3a-2 failed as registered,** but the *opposite* direction appeared in all three samples: counter value is higher when the opponent had advanced deeper before losing the ball.
- **Weak spots: no.** Opponent concession profiles by channel (build-up, counter, set pieces…) were not persistent in Wyscout. A matchup model beat main effects out of sample only because *attackers'* specialisations persist; the defender "weak-spot" term was null in both datasets (+0.07 ± 0.06, −0.03 ± 0.05). Scouting an opponent's channel weakness, at this level of detail, does not predict where they will concede.

Round 4: defensive structure from shot freeze-frames

Every StatsBomb shot records where all players stood.

- The number of defenders inside the shooter–posts triangle per shot conceded ("cone density") is a highly persistent team trait (reliability 0.75) and unrelated to team rating ($r = 0.07$). A coachable, unbought trait, apparently.
- Team-level, it went with conceding fewer goals than the market expected (−1.06 goals/match per extra defender, $p = 0.026$).
- Conceding below xG also repeated (reliability 0.39–0.45), but a decomposition shows that is mostly goalkeeping, and partly wealth-linked.

Round 5: the cone-density candidate dies

Three pre-registered confirmations, all of which **failed**:

- prospective (first-half trait → second-half goals vs market): right sign, $p = 0.28$;
- unseen leagues (WSL 2018–21, NWSL 2023, ISL 2021/22; 577 matches): wrong sign;
- shot level in the unseen leagues: $p = 0.10$.

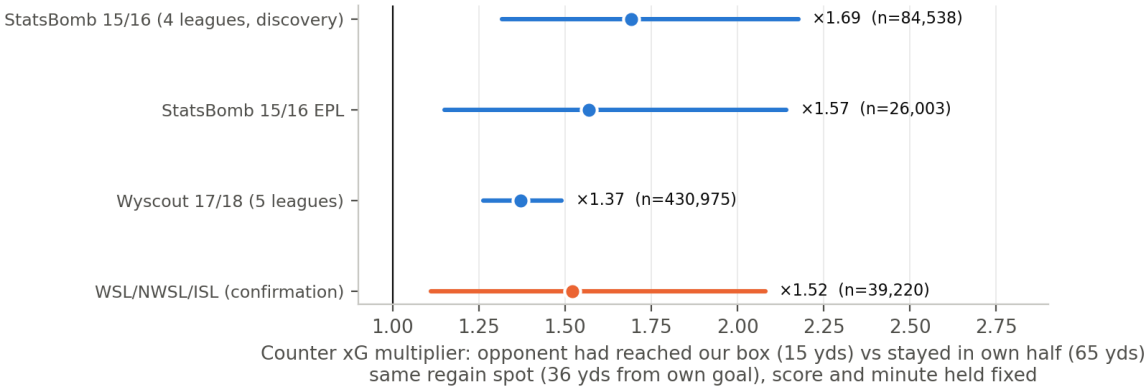
The round-4 association was noise or confounding (most plausibly goalkeeper quality). Without the confirmation step it would have been our headline.

Round 6: the two survivors, confirmed on unseen leagues

The same 577 unseen matches. Team strength was measured as leave-this-match-out xG for and against, since these leagues have no odds.

Pre-registered test	Result	Verdict
H6-1 Set-piece xG is less strength-elastic than open play	0.72 vs 1.02; difference 0.30, team-bootstrap CI [0.11, 0.50]; same direction in all 5 competitions	**Confirmed**
H6-2 Underdogs keep a larger share of set-piece xG in mismatches	29.1% vs 26.8%, CI [−0.9, +5.7] pp	Not confirmed (right sign, n = 330)
H6-3 Counter value rises when the opponent had pushed deep	marginal effect −1.01 (SE 0.38), one-sided p = 0.004	**Confirmed**

Fig 9. 'Break after they commit': the effect holds in four independent populations



How to read "break after they commit." Hold the place where you win the ball (here 36 yards from your own goal), the score and the minute fixed. If the opponent had just been inside your box, the possession you start is worth about 1.5× the counter xG it would be worth had they never left their own half. Their players are committed forward and out of position. The effect sits in the 1.4–1.7 range in men's and women's football, two providers and three seasons.

It is an **in-the-moment decision rule**, not a style. Teams that defend deeper do win more of their balls in this window (+4 pp of regains per SD of block depth, $p < 10^{-12}$). But they do not create more counter xG per match, because those regains happen nearer their own goal (`results/r6_bait.json`). So the lever is *what you do with the ball* after the opponent has over-committed: release runners and play forward immediately. Sitting deeper to provoke it is not the lever.

9. Part III: three more "football is doing it wrong" hypotheses that died

Round 7: do teams chase too late? (football's "pull the goalie" question) * Trailing teams barely escalate: own xG rises only +10–12% from 45–60' to 75'+, and the CIs include 0 in all three populations. * Late in matches a trailing team also *concedes less* (×0.67–0.87), because the leader stops attacking. Any "go all-in earlier" policy built on late-game rates therefore borrows the opponent's passivity. That is the confound we registered in advance, and it invalidates the timing optimisation.

Round 8: the dead-ball decision audit (338k corners, free kicks and throw-ins). * Discovery flagged two "misallocations": corners delivered to the penalty-spot area (×1.41 xG vs near-post) and long throws into the box (×1.67 vs short). * Both **confirmed on the letter** in all three held-out populations (×1.18 and ×2.25, $p < 0.0001$). * Both **dissolved** under the selection checks we had registered. Compared at the same throw

location, long throws are worth nothing extra (−0.18, +0.27, +0.04, −0.01 log-points). For corners, comparing each team with itself flips the discovery sign. * This is the clearest demonstration in the project of how convincing, replicated, *wrong* tactical findings arise.

Round 9: referees as a natural experiment. Teams do not choose the referee, so strictness is quasi-random. Over 7,000 EPL matches, referee strictness had no effect on underdog results against the market (discovery +0.011 ± 0.024, confirmation +0.024 ± 0.030), no interaction with underdog fouling, and no effect on total goals.

Fig 8. 28 pre-registered tests, nine rounds: 7 held

R1	P1 style predicts next-segment xG	✗
R1	P2 in-match policy ≥0.05 pts (cross-fit)	✗
R1	P3 habitual style beats market	✗
R1	P4 high-tempo underdogs lose vs market	✓ **held**
R1	P5 risk-seeking raises title odds	✗
R2	3 box-score traits/matchups (BH-selected)	✗
R2	ML on 22 style traits beats market	✗
R3	Opp. commitment improves regain model	✗
R3	Counter value falls if opp. was deep	✗
R3	Top 20% regains carry ≥40% of xG	✓ **held**
R3	Concession profiles persistent	✗
R3	Matchup model beats main effects*	✓ **held**
R4	Structure predicts goals beyond xG	✗
R4	Cone density persistent, not bought	✓ **held**
R4	Over-performance vs xG repeats	✓ **held**
R5	Cone density → fewer goals (prospective)	✗
R5	… on unseen leagues	✗
R5	… shot level, unseen	✗
R6	Set pieces flatter than open play (unseen)	✓ **held**
R6	Underdog keeps more set-piece xG (unseen)	✗
R6	Counter value rises if opp. was deep (unseen)	✓ **held**
R7	Trailing teams escalate late	✗
R7	Teams chase too late ('pull the goalie')	✗
R8	Corners to penalty spot beat near post†	✗
R8	Long throws into box beat short†	✗
R9	Strict referees hurt underdogs	✗
R9	… especially foul-heavy underdogs	✗
R9	Referee strictness shifts total goals	✗

† passed the registered confirmation, then failed the registered selection checks.

10. Part IV: why the graveyard, and the way out

10.1 The power problem

From the observed outcome variances we computed the number of decisions needed to detect a tactic with 80% power (`results/r10_power.json`).

A tactic that improves…	by	needs	= seasons of one club	= seasons of a 20-club league
corners (xG per corner)	20%	2,823 corners	14.9	0.74
attacking free kicks	20%	1,817	15.9	0.80
attacking throw-ins	20%	11,740	34.3	1.7
transitions (xG per regain)	20%	20,335 regains	7.4	0.37
the match (points vs market)	+0.1 pts/match	2,176 matches	57	2.9

A tactic worth four points a season, enough to change a mid-table club's year, is undetectable in a decade of its own results. Almost all of the variance in a decision's outcome sits *within* a match (72–98%), so pairing decisions within matches barely helps. This is why observational mining fails: real effects are too small relative to noise, and anything large enough to see is usually selection.

10.2 Surrogate endpoints

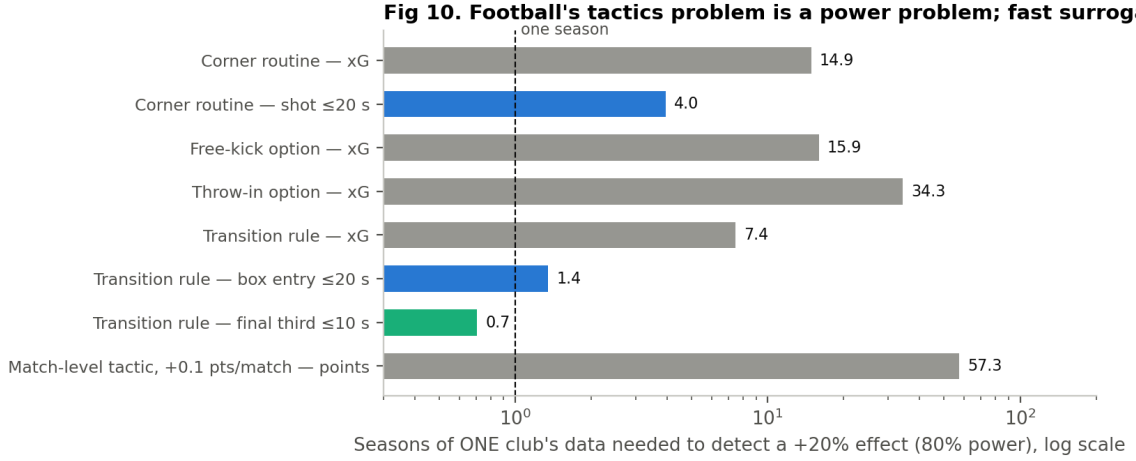
Medicine's answer to rare outcomes is to measure a frequent outcome that sits on the causal path. For transitions:

Surrogate after a regain (not already in the final third)	Rate	Share of regain xG	Team-level validity*	Club-seasons for +20%
xG itself	—	100%	0.31	7.4
shot within 20 s	4.9%	99%	0.17	3.4
box entry within 20 s	11.7%	87%	0.28	1.4
final third within 10 s	**20.2%**	**66%**	**0.27**	**0.7**

*Correlation of the surrogate rate in a team's odd matches with its xG per regain in its even matches.

For corners, "won the first contact" fails as a surrogate (validity −0.19). Corner experiments need multi-club pooling.

Fig 10. Football's tactics problem is a power problem; fast surrogates fix part of it



The confirmed vulnerability effect is visible in the surrogate. After the opponent had pushed into the box, the odds of reaching the final third within 10 s are 1.47× higher [1.36, 1.59] in StatsBomb 2015/16, and 1.26× [1.08, 1.47] in the unseen leagues (`results/r10_window_surrogate.json`).

10.3 A trial a single club could run this season

The Underdog Transition Trial. * **Intervention:** a "go rule". When the ball is won in our own half within ~5 s of the opponent having been inside our final third, at least two players sprint forward and the first pass must be forward. * **Control:** the club's normal transition behaviour. * **Randomisation:** by match (or by half, with a pre-match signal), stratified by opponent strength. * **Primary endpoint:** final-third entry within 10 s per qualifying regain. * **Secondary endpoints:** xG within 20 s, and counter xG conceded within 30 s (the cost side, which history cannot measure).

Sample size, from the data: a club has about 21 qualifying regains per match (won in its own half after the opponent had been in its final third). In these, the 10-second final-third rate is 10.9%. Detecting a relative uplift of +20% / +30% / +50% needs about 3,200 / 1,420 / 510 qualifying regains. That is **4.0 / 1.8 / 0.6 seasons** for one club, or a third of that for a three-club consortium. Historically, the deep-vs-shallow contrast moves the surrogate's odds by +26–47%. An intervention that captures part of that is detectable by a single club in about two seasons, or by a small consortium in under one. A *general* transition rule applied to all regains needs only 0.7 club-seasons for +20% (§10.2).

11. What this means

1. **What a weaker team can use today:** dead balls, where quality matters a third to two-thirds as much as in open play, and the vulnerability window ("if they were in our box when we won it, go"). Both were confirmed in four populations. Together with the round-1 levers they are worth a few points a season: Brentford's top-four odds move from 3% to 5–9%, and title odds stay at or below 0.15%.
2. **What does not work, measured:** steering matches with in-game style changes; out-building better teams; scouting channel weak spots; risk-seeking for titles; chasing earlier; long-throw and corner-target "hacks"; adapting to referees. Neither box-score style nor style matchups beat the market after 2017.
3. **Why football has not had its data revolution, and how it could:** tactical effects worth points are an order of magnitude below what any club's history can detect. Observational mining therefore produces confident false discoveries. In this project, four candidates passed at least one stage and then died. The fix is not more data mining or bigger AI simulators. It is **randomised tactical trials with validated surrogate endpoints**, which make a single-club, single-season test feasible for transition tactics.

4. **The sports-changing claim we are prepared to make:** the first club that runs tactics as experiments, not as opinions or as mined history, gains a learning speed no rival can match from data alone. Our trial protocol shows that this is feasible now.

12. Limitations

- **Identification.** Every frontier is observational. Lagging, game-state control, like-for-like control groups and cross-fitting reduce but cannot remove confounding. A coach-run controlled trial (milestone 4 in the brief) is the only clean test.
- **Unseen-league strength.** WSL/NWSL/ISL have no odds, so round 6 uses leave-one-out xG strength. That is noisier, and it makes set-piece flatness harder to find, not easier.
- **Event data, not tracking.** Pressing structure, defensive line height and off-ball runs are only proxied by on-ball events. The levers we could not measure may be the ones that matter. Licensed tracking data (e.g. 2025/26 Premier League) is the obvious next step and could overturn P1/P2.
- **Quality measure.** Market odds include the effect of a team's habitual style, so "wealth correlation" uses a rating that is not purely about player quality. Transfermarkt squad values were not accessible to our pipeline.
- **Single seasons.** Cheap-skill reliabilities come from one season per dataset (81 and 98 team-seasons) and differ between datasets.
- **Season model.** Brentford's set-piece baseline is assumed median. If they are already above median (likely), the headroom is smaller than shown.
- **Surrogate validity is observational.** Team-level predictive validity does not guarantee that an intervention moving the surrogate moves goals. The trial must track xG as a secondary endpoint, and pooled trials must validate the surrogate.
- **Pre-registration wording.** P2's literal criterion could be met by noise. We flag this rather than claim it.

13. Reproducibility

Everything runs from `underdog/src/` in this order: `fetch_statsbomb.py`, `odds.py`, `build_segments.py`, `p4_tempo_market.py`, `p4_robust.py`, `p1_heldout.py`, `tactical_changes.py`, `event_study.py`, `p2_policy.py`, `possessions.py`, `channels.py`, `p3_style_market.py`, `theory_vs_p4.py`, `xg_model.py`, `wyscout_replication.py`, `season_sim.py`, `season_sensitivity.py`, `p5_and_city.py`, `leicester.py`, `setpiece_market.py`, then rounds 2–10: `discovery.py search|confirm|ml`, `r3_regains.py build|test`, `r3_weakspots.py`, `r4_structure.py`, `r4_followup.py`, `r4_value.py`, `r5_confirm.py`, `r6_confirm.py`, `r6_bait.py`, `r6_vuln_all.py`, `r7_segments.py`, `r7_chase.py`, `r8_deadballs.py build|audit`, `r8_robust.py`, `r9_referees.py discovery|confirm`, `r10_power.py`, `r10_surrogates.py`, `r10_window_surrogate.py`, and finally `make_figures.py`. Every number in this paper is in `underdog/results/*.json`. The seed for all bootstraps is 20261011.

Data: StatsBomb Open Data (attribution required); Pappalardo et al. (2019), "A public data set of spatio-temporal match events in soccer competitions", Scientific Data 6:236; football-data.co.uk.